atomQUEST

2025

Problem Statement

Create a **robotic cleaning bot** that is versatile, efficient, and smart. The bot can be equipped with various cleaning mechanisms such as **brooms, vacuum suction, liquid floor cleaners**, and more.

The bot will be tested in a **challenge room** that contains **pre-defined but randomly placed obstacles** and **various types of dry garbage** including dust, paper, wire strands, wrappers, hair, etc., spread across the floor.

The bot that **cleans the maximum amount of dust within the allotted time** will be declared the winner!

Competition Structure

Round 1: Abstract Submission

- **Deadline**: 20th August 2025
- Participants must submit two/three pager (excluding intro) an abstract containing at least the following
    - Team members details
    - List of team members (team needs to contain at least 3 members)
    - Institute details
    - List of labs and facilities at your institute that will help you realize the prototype
    - While having a faculty mentor is not mandatory at this stage, it is encouraged. If you have identified faculty mentor(s), please mention their brief bio.
    - Rough list of components (electronic and mechanical) for your bot
    - System level block diagram
        - Rough specifications of your bot
        - Battery sizing
        - Number and specs of motors
        - Approx. Weight of the system
        - Details of the controller
        - Details of the sensors
        - Chassis material
        - Number and specs of wheels
        - Dust storage capacity
        - Cleaning mechanism
        - Storage container sealing
        - Removal and reattach mechanism for the storage container

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Round 2: Proto Submission- Level 1

- **Deadline**: 12th September 2025
- Submit a 90-second video showcasing:
    - Final team members
    - Overview of the idea
    - Components procured so far
    - Pending components
    - Planned next steps

- Also include a signed letter from your **faculty mentor** confirming their support until the finale.

Round 3: Proto Submission- Level 2

- Submit a 90-second video showing:
  - The bot in motion
  - The bot cleaning some dust
  - (Optional) Any extra feature demonstrated
- Additionally, submit:
  - A one-pager on **battery life calculation**
  - A one-pager on **cost calculation**

You can complement the video with CAD designs and/or a 2-pager document describing your prototype.

Mentorship

- Shortlisted teams from Round 3 will be paired with Atomberg mentors, who will guide them through the development phase, providing insights, feedback, and technical support.
- Teams can use these mentorship sessions to refine their solutions and overcome challenges.

Active participation during the mentorship phase is mandatory.

Round 4: Grand Finale

- The **Grand Finale** will be held **in person** at Atomberg Innovation Center. Finalists will present their **fully developed solutions** to Atomberg's leadership team and a jury of industry leaders.
- Teams face off in a high-stakes war room where talent, strategy, and innovation collide.
- Details of the Grand Finale will be shared closer to the event.

Finale Day Setup & Evaluation

What You'll Know in Advance

- Shape and dimensions of the challenge room
- Nature and number of obstacles
- Types of garbage

What Will Be Unknown

- The exact **placement of garbage and obstacles** — they will be randomized for every team
- Each challenge room setup will be **unique** for every team pairing

Allowed Dimensions of the Bot

- **Length**: max 1.5 ft
- **Breadth**: max 1.5 ft
- **Height**: max 1 ft